

AI-Assisted Event-Driven Application for Small and Medium-Sized Businesses

Background & Overview: Small and medium-sized enterprises (SMEs) in emerging markets rely heavily on informal, conversational order-taking pipelines (such as chat notifications and voice notes) alongside physical paper receipts. While highly flexible, these informal methods fundamentally fail to scale, creating operational disconnects with critical backend inventory ledger adjustments, billing transactions, and real-time operational analytics. To bridge this computational gap without forcing behavioral changes, this project proposes an AI-Assisted, Event-Driven platform. It translates conversational text and digitized invoices into structured business records, backed by an append-only event log, and orchestrates decoupled, asynchronous business services.

This document delineates the core engineering goals and system architecture. The checklist below defines the specific design parameters and progress milestones. The subsequent sections zoom in on the modules that implement each objective.

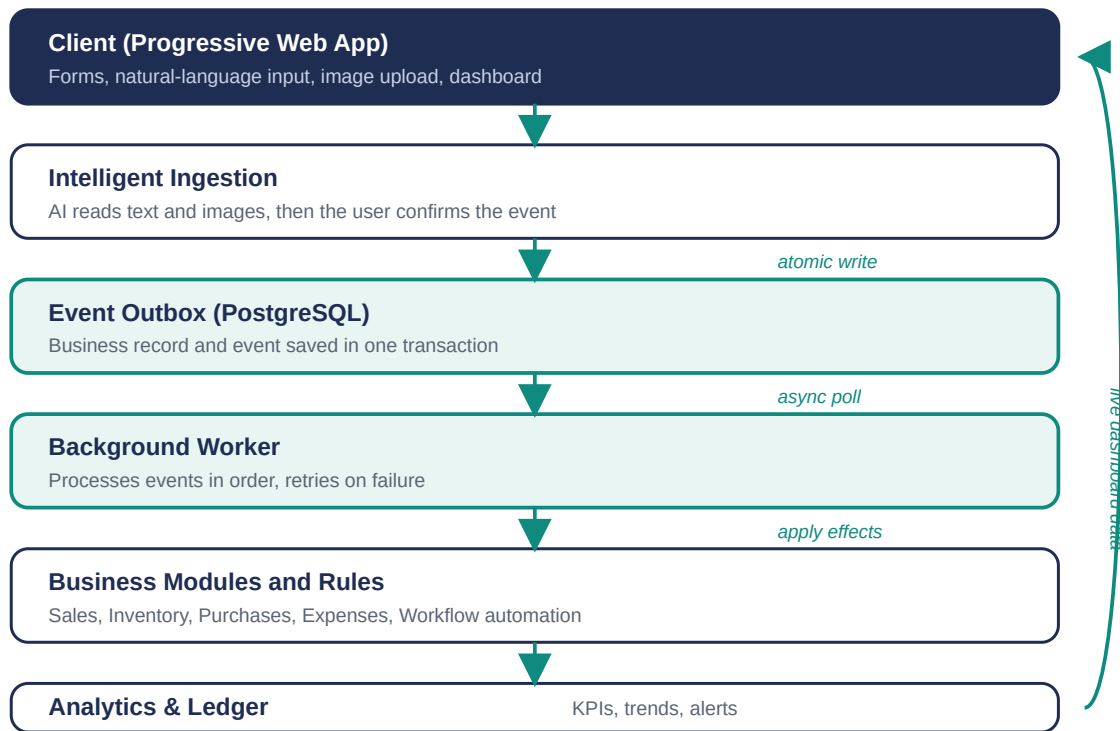
Objective / Project Milestone	Syllabus Scope & Technical Description	Status
1. Intelligent Input Ingestion	Automate structured information extraction from conversational plain-text notes and photographed business receipts using multi-modal AI.	
2. Asynchronous Event Bus	Implement a reliable server-side event queue using the Transactional Outbox pattern on PostgreSQL to decouple core business writes from background processing.	
3. Modular ERP Core Services	Model the main business services (Sales, Inventory, Billing, Analytics) as decoupled, normalized relational schemas bounded to multi-tenant boundaries.	
4. Configurable Automation	Build a declarative rules engine that runs background actions (stock alerts, expense flagging, notifications) from event triggers, configurable without code changes.	
5. Dashboard & Evaluation	Deliver a mobile-first Progressive Web App to visualize transactional records, and measure extraction accuracy and event propagation latency.	

Project Team Members

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System Architectural Blueprint

Intended system design: The system is layered and event-driven. Rather than adding a separate message broker, it uses one database as both the system of record and the event queue: each confirmed action saves its business record and its event together, and a background worker then processes events asynchronously so the interface stays responsive. The client is a mobile-first web app, while the database and worker run on the server. The diagram below shows the flow from a user's input through to the stored records and the dashboard.



End-to-end flow: input is confirmed, then written and processed as events; results feed the dashboard.

Objective-Wise Functional Focus

Objective 1: Intelligent Ingestion

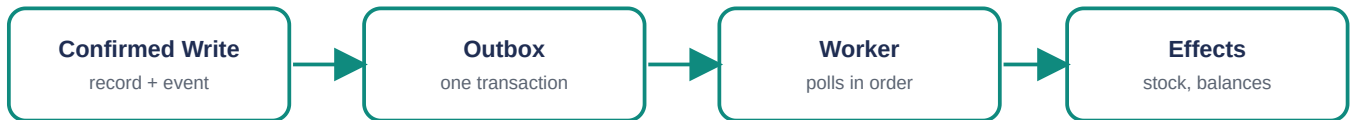
Turns informal input, both typed notes and photographed receipts, into a single structured business event. A parsing step proposes the event, and the user reviews and confirms it before anything is saved, so a wrong reading never reaches the records.



Objective-Wise Functional Focus (Continued)

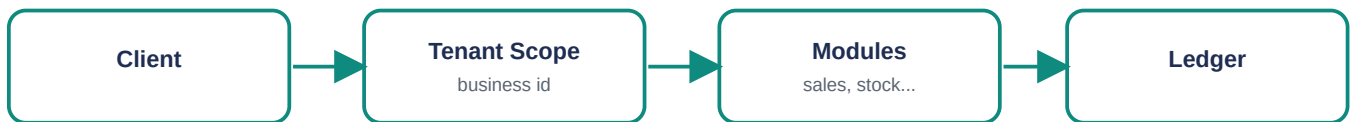
Objective 2: Asynchronous Event-Driven Coordination

Every confirmed action writes its business record and its event together, in one atomic step, so an event is never lost and never emitted for a change that did not happen. A background worker then processes events one at a time, safely and in order, applying their effects and retrying if something fails.



Objective 3: Tenant-Isolated Modular Business Functions

Business features (sales, inventory, purchases, expenses, analytics) are separate modules over a shared database. Every record is tagged to the business that owns it, so one tenant's data stays fully isolated from another's.



Objective 4: Configurable Automation Rules Engine

Routine responses are handled by configurable rules rather than hard-coded logic. Each rule watches for an event, checks a condition, and runs an action such as a low-stock alert or flagging a large expense. Rules are edited in the app, so behavior can change without touching code.



Objective 5: Consolidated Dashboard and Evaluation

A mobile-first dashboard brings the live state together as totals, trends, and alerts. The project is evaluated on how accurately it turns informal input into correct events, and how quickly those events propagate to the records.

